SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)
Constituent of Symbiosis International (Deemed University), Pune
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॥वसुधैव कुटुम्बकम्॥ Re-Accredited by NAAC with 'A' Grade

Department of Computer Science and Engineering
DevOps Lab - L3

Lab Assignment No: 13	
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Batch	2023 - 2027
Class	CSE - B1
Academic Year & Semester	2026 - 2027, 7th Semester
Title of Assignment	Kubernetes Service & Networking

Implementation & Theory	```yaml
	apiVersion: v1
	kind: Service
	metadata:
	  name: web-app-service
	  labels:
	  app: web-app
	spec:
	  selector:
	  app: web-app
	  ports:
	  - port: 80
	  targetPort: 5000
	  type: NodePort
	``
	Service YAML created — Wrote web-service.yaml defining a Service that selects Pods labeled app: web-app (matching the Deployment from assignment 12), listening on port 80 internally and forwarding to targetPort: 5000 (the Flask app's container port), with type: NodePort to expose it outside the cluster.

```
service/web-app-service created
(base) raghavdhoot@Raghavs-MacBook-Air-2 k8s-manifests %
```

Applied the manifest — kubectl apply -f web-service.yaml created the Service successfully.

```
NAME              TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)          AGE
kubernetes        ClusterIP   10.96.0.1       <none>        443/TCP          9m25s
web-app-service   NodePort    10.107.84.70    <none>        80:31547/TCP     30s
(base) raghavdhoot@Raghavs-MacBook-Air-2 k8s-manifests %
```

Verified Service details — kubectl get service showed web-app-service as type NodePort, with Kubernetes auto-assigning a random high port (e.g. 80:3XXXX/TCP) mapped to the internal port 80.

```
|-----------|-----------------|-------------|------------------------|
| NAMESPACE |      NAME       | TARGET PORT |          URL           |
|-----------|-----------------|-------------|------------------------|
| default   | web-app-service | 80          | http://192.168.49.2:31547 |
|-----------|-----------------|-------------|------------------------|
🔗  Starting tunnel for service web-app-service.
|-----------|-----------------|-------------|------------------------|
| NAMESPACE |      NAME       | TARGET PORT |          URL           |
|-----------|-----------------|-------------|------------------------|
| default   | web-app-service |             | http://127.0.0.1:60377 |
|-----------|-----------------|-------------|------------------------|
🎉  Opening service default/web-app-service in default browser...
❗  Because you are using a Docker driver on darwin, the terminal needs to be open to run it.
```

Hello, Docker!
Log:

```
2026-10-12 13:02:59: Hello, Docker!
```

Accessed the application — Used minikube service web-app-service to resolve the Minikube node IP and NodePort, which opened the app in the browser confirming "Hello, Docker!" — proving traffic was correctly routed from outside the cluster, through the NodePort Service, and load-balanced across the two Deployment Pods.